

Education

Indian Institute of Technology (IIT) Goa	In Progress	M.Tech in VLSI and Signal Processing	2024 – Present
Sinhgad Institute of Technology, Lonavala	CGPA: 7.64	B.E. Electronics & Telecommunication	2017 – 2021

Academic Research

Master's Thesis: Smart Periodontal Probe Design with Embedded Sensors	
<i>Guide: Dr. Bidhan Pramanick, IIT Goa</i>	2024–Present
- Designing smart periodontal probe with integrated biomedical sensors. - Implemented signal processing for accurate gum depth/tissue response. - Modeled in COMSOL; focus on low-power and ergonomic design.	

Teaching Assistantship

Teaching Assistant: Data Communication Lab	
<i>Guide: Dr. Neelakandan R, IIT Goa</i>	Jul 2025 – Present
- Guided 40+ students in data communication and error detection experiments - Reduced experiment downtime by 25% via troubleshooting support. - Streamlined lab operations with proactive equipment setup.	

Internships

Full-Stack Data Engineering Intern	Expertshub, Allahabad, 2021
- Built predictive data models & sentiment analysis pipelines in Python, improving accuracy by 5 - Led 2-member team to streamline deployment process. [Certificate]	
Cloud & Web Development Intern	Techmaniacs EduServices, Lonavala, India, 2021
- Developed web applications and improved UI design using HTML5 - Built and deployed scalable cloud-based services on AWS using Python, reducing deployment overhead by 10 - Delivered a comprehensive final report with full documentation [Certificate]	
IoT Intern	Expertshub, Bangalore, India, 2021
- Designed IoT applications integrating multiple sensors and microcontrollers for real-time data collection. - Engineered a Google Assistant-controlled robot, enabling seamless voice-to-action execution via IFTTT. - Authored detailed documentation, improving reproducibility and project handover efficiency. [Certificate]	

Projects

Two-Stage Operational Amplifier (Cadence)	
<i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2025
Designed and simulated a two-stage op-amp (65nm PDK) in Cadence Virtuoso; validated with DC, AC, and transient analyses [Report].	
4-bit Adder Design & Delay Analysis (Electric VLSI)	
<i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2025
- Implemented a ripple-carry adder; verified functionality and timing in LTspice. - Analyzed propagation delays; critical path delay = 547.32 ps [Report].	
Energy Management System (FSM, VHDL)	
<i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2024
Developed a finite-state machine for dynamic energy load management using Quartus and ModelSim [Paper].	
NMOS Differential Amplifier (Electric VLSI)	
<i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2024
Implemented and analyzed an NMOS differential amplifier; measured gain and output swing in LTspice [Report].	
Audio Equalizer (DSP Kit)	
<i>Guide: Dr. Sujit Kumar Sahoo</i>	2024
Built a real-time Chebyshev filter-based audio equalizer on a TI DSP kit; tested using Keil µVision5.[Code]	

IoT-based Health Monitoring Kit (Final Year Project)

2021

- Developed an IoT-based kit integrating vital-sign sensors and cloud monitoring; showcased at TECHNOVATION 2K21. [Presentation], [Video], [Certificate].

Additional Projects

Drone Project (2018)

SIT, Lonavala

- Engineered Arduino Uno-based quadcopter for land tracking and mapping.
- Programmed predefined flight trajectories for stable navigation. [Photo].

Virtual Assistant "Christopher" (2019)

SIT, Lonavala

- Developed Linux-based virtual assistant for routine task automation. [Video Clip].

Volunteer Experience

Speaker, IIT Goa – Conducted PCB Design & Soldering Workshop (Eagle). [Certificate] 2025

Seminar Mentor, IIT Goa – Inspired PM Shri KV students with drones & embedded demos [Photo] 2025

IoT Mentor – Guided Smart Mirror project (SIT, 2020) [Certificate] 2020

Skills

Programming VHDL, Verilog, Python, C/C++, Java, MATLAB

Tools Cadence, LTspice, Electric VLSI, Quartus, Arduino IDE, COMSOL Multiphysics, Autodesk Eagle

Domains VLSI Design, DSP, Embedded Systems, Biomedical Sensors, IoT Healthcare

Cloud AWS, Google Cloud, Intel OpenVINO Toolkit

Certifications

- Advanced Drone Technology (Air Taxi) Workshop – Bharat Space Education Research Centre & I-STEM, hosted at IIT Goa, 2025 [Certificate]
- MATLAB Onramp (Mathworks, 2021) [Certificate]
- Google Cloud Certified Professional Cloud Architect (Coursera, 2020) [Certificate]
- Neural Network from Scratch in TensorFlow (Coursera, 2020) [Certificate]
- Intel OpenVINO Toolkit for Computer Vision (Coursera, 2020) [Certificate]
- Fundamentals of Parallelism on Intel Architecture (Coursera, 2020) [Certificate]
- ISRO Online Learning – "Application of Geoinformatics in Ecological Studies" (ISRO, 2020) [Certificate]
- Microprocessor & Microcontroller Certifications (Bharat Acharya Education): 8085 (2023) [Certificate] | ARM (2020) [Certificate] | 8051 (2020) [Certificate] | 8086 (2020) [Certificate] | COA (2020) [Certificate]

Achievements

Model United Nations Represented Panama at BITS Goa MUN (UNSC), debated arms trade & counter terrorism [Certificate] 2025

GATE EC Scored 340/1000, All India Rank (AIR) 9467 [Scorecard] 2024

TCS iON NQT Cognitive: 1101.32/1800 (61.18%) and Psychometric Assessment [Scorecard] | [Report] 2023

TECHNOVATION 2K21 State Level Project Presentation [Certificate] 2021

Elocution Competition, SIT Lonavala 1st Prize [Certificate] 2021

Model Mania Exhibition Won against all SIT colleges; 1st Prize, Sinhgad Tech Fest, Prize: 10,000 [Certificate] 2019

Bytes of India Feature Drone Project showcased and recognized [Article] 2019

Positions of Responsibility

PG General Secretary, IIT Goa Elected liaison for PG students, 2025–26 2025

Department Representative, Karandak (Sinhgad Intercollegiate Fest) Led a 10-member core team to organize 12 events, boosting participation by 15% [Certificate] 2020